CODE:

```
% Facts

fact(a).

fact(b).

% Rules

rule(c,[a,b]).

rule(d,[c]).

rule(e,[d]).

% Backward Chaining

prove(X) :-

  fact(X).

prove(X) :-

  rule(X,Conditions),

  prove_all(Conditions).

prove_all([]).

prove_all([H|T]) :-

  prove(H),

  prove_all(T).
```

OUTPUT:

```
56 ?- consult(['c:/Users/arunp/OneDrive/Documents/AI/EXP 30.pl']).
Warning: c:/users/arunp/onedrive/documents/ai/exp_30.pl:2:
Warning:    Redefined static procedure fact/1
Warning:    Previously defined at c:/users/arunp/onedrive/documents/ai/exp_29.pl
:4
Warning: c:/users/arunp/onedrive/documents/ai/exp_30.pl:6:
Warning:    Redefined static procedure rule/2
Warning:    Previously defined at c:/users/arunp/onedrive/documents/ai/exp_29.pl
:8
true.

57 ?-  prove(e).
true .

58 ?- prove(d).
true
```